

Generative model closure checklist (SCOUT C1–C8)

Canonical name: `06_Generative_closure_checklist.md`
Original archive: `obsolete/original_filenames/20260615_1012_SCOUT_to_COMPASS_minimal_model_closure.md`
Date: 2026-06-15 (checklist); D10 export done 2026-06-24 — see `08_Basketball_figures_on_disk.md`
Charles: Path II unchanged. **C1–C4 + C7 → green** after D10 bundle. C6 partial (prediction #2 still Army/CODA). C8 exported in bundle.

Note: Body below retains June 15 “pre-D10” color counts in places; trust the **June 24** status line above and `08_Basketball_figures_on_disk.md` for what is on disk today.

§ ambiguity: Headings **§1–§6 below** are subsections **of this checklist file**. Manuscript sections use different numbers — see `#01` **Symbol systems**.

Summary (one screen for Alex)

Color	Count	Meaning
Green	3 now → 5 after D10	Scientifically demonstrated; needs packaging only
Yellow	4 now → 2 after D10	Done in notebook/playground; not yet frozen for manuscript
Red	0	Nothing missing that blocks manuscript §5 (generative) draft under Path II

Single next task: D10 manuscript export bundle (`export_scout_manuscript_bundle_v1.py`) — **1–2 sessions**, packaging only.

§1 — Closure checklist (SCOUT-owned)

#	Criterion	Status	Evidence	Blocker	Color
C1	Talent-only (ability-only) generative baseline fails qualitatively vs inverted-U stylized fact	DONE	538D CELL 10: <code>w=0</code> / ability-only score → monotone on pool-mean axis; documented in <code>538_Cell10_Generative_Manual.md</code>	Export to bundle (D10)	Yellow → Green after D10
C2	Congestion-in-score run produces stable qualitative peak-and-decline (pool-mean axis OK under Path II)	DONE	538D CELL 10 + <code>SELECTION_539_*</code> preset (<code>tier1_sim_config.py</code>); state in <code>sports/tier1_cell10_playground_state.json</code>	Export PNG + settings JSON to bundle	Yellow → Green after D10
C3	Axis table frozen: model quantity ↔ empirical quantity ↔ setting	PARTIAL	Prose + two-row logic in <code>20260611_1640_SCOUT_to_COMPASS_model_coherence.md</code> §A3; no standalone <code>axis_table_generative_readouts.md</code> on disk yet	SCOUT D10	Yellow → Green after D10
C4	Score equation one-pager frozen with λ / soft-assignment semantics	PARTIAL	<code>sports/tier1_sim_config.py</code> (<code>SELECTION_539_*</code> constants); <code>538_Cell10_Generative_Manual.md</code>	SCOUT: extract to <code>score_equation_one_pager.md</code> in bundle	Yellow → Green after D10
C5	Honest limitation prose drafted (Rung 2 axis ≠ Rung 1 axis)	DONE	Exact VECTOR sentence in <code>20260611_1640_SCOUT_to_COMPASS_model_coherence.md</code> §A3	VECTOR ink	Green (prose ready; VECTOR pastes)
C6	≥ two predictions traceable to mechanism (not curve replication); includes model-guided measurable (quality vs congestion)	PARTIAL	(1) Near-threshold heterogeneity: <code>datasets/mbb/exports_inverted_u_v0/heterogeneity_ventiles_top_tail.{png,csv}</code> (538D CELL 4D, 2026-06-02). (2) Peak shift with global Λ: not basketball-owned — Army/CODA natural home; basketball K-slots secondary	Prediction #2 = CODA + VECTOR prose; #1 export refresh optional in D10	Yellow (1 of 2 strong in SCOUT lane)
C7	Manuscript export bundle on disk (D10): empirical Fig 2 + generative contrast + Tier 2.5 feature panel + manifest	MISSING	Script specified in <code>20260611_1640_SCOUT_to_COMPASS_model_coherence.md</code> §D; not yet built	SCOUT D10; Charles C9 path lock	Yellow until built
C8 <i>(PD12 optional)</i>	Quality vs congestion distinction exported in D10 (axis table row + congestion column/panel)	PARTIAL	<code>crowding_smooth</code> in pipeline; not yet named closure row on disk	SCOUT D10	Yellow → Green after D10

SCOUT closure rule: Items C1–C4 + C7 go **green** when D10 lands. C5 already green. C6 stays **yellow** until cross-domain prediction map names Army Λ + basketball near-threshold — acceptable for v1 **manuscript §5** (generative) closure; **manuscript §7** (predictions) can carry yellow honestly.

§2 — Honest inventory (blunt, per rung)

Rung 1 — Empirical inverted-U on LOO pool quality

Publication-ready for basketball with one housekeeping refresh. The inverted-U on `poolq_loo` is established in `sports/530_sports_pipeline.ipynb` / `sports/538_alex_tier1_model_and_fit.ipynb`; best frozen artifacts on disk are April 2026 ventile exports (`inverted_u_ventiles_ppm_zwithinseason_2026-04-06.png` + CSV under `datasets/mbb/exports_inverted_u_v0/`). **Single refresh:** re-run `530` export with a June dated slug before VECTOR locks Figure 2 captions — science unchanged, provenance hygiene only. No generative machinery required for Rung 1.

Rung 2 — Minimal generative mechanism (`sports/538D_development.ipynb` CELL 10)

Demonstrated interactively, not yet manuscript-frozen. Score $S_i = A_i - \lambda L_{C,LOO}$ with soft assignment runs in 538D CELL 10; 539 preset loads `SELECTION_539_*`; talent-only fails; congestion-in-score shows peak-and-decline on **pool mean**. **Playground-only today:** generative PNGs live in widget output + `tier1_cell10_playground_state.json`, not in `exports_inverted_u_v0/`. **Exportable in one session** via D10 script.

Rung 3 — Predictions / decomposition

One prediction artifact exists on disk; one is cross-domain. Near-threshold heterogeneity (`sports/538D_development.ipynb` CELL 4D) exported June 2026. Mean×SD (CELL 4B/4C) and full Wang-ladder LPM exports remain notebook-only — supplement tier, not closure blockers. Peak-shift-with- Λ is conceptual in basketball; CODA owns Army instantiation.

§3 — Explicit non-requirements (v1 — stop worrying)

SCOUT confirms COMPASS draft list. Charles should ignore until post-draft:

Item	SCOUT verdict
Generative LOO-pool-quality bin-for-bin match	Out of scope v1
Per-domain calibration of λ (Army / sports / tenure)	Out of scope v1
Full deconstructable B–D estimation in generative simulation	Out of scope v1
Mean × dispersion as primary prediction	Deferred (VECTOR)
Network extensions as closure criteria	Out of scope v1
SCOUT draft Cox / tenure Layer B	Out of scope v1

SCOUT additions:

- **538D decomposes 539** as a manuscript claim — false; do not require before draft.
- **Generative match on LOO pool quality axis** even qualitatively — not closure (honest limitation panel only).
- **June-dated empirical Fig 2** — nice-to-have before submission, not Tier 2 closure.
- **Time-to-draft Cox / competing risks in basketball** — Army-only sophistication (CODA Q11; SCOUT agrees).
- **Harmonized bin count across settings** — PEER deferred (18 equal-width tenure bins vs basketball ventiles); cosmetic only.
- **Frozen generative parameter sensitivity sweep** — post-draft robustness.

§4 — Cross-domain: basketball Tier 2 closure

Setting	Basketball generative enough for Tier 2 closure?	What this setting must show instead
Army (CODA)	Yes — Army does not need generative sim for v1	Empirical Rung 1: CIF inverted-U on LOO pool minus mean + cause-specific Cox with quadratics. Predictions: near-threshold effects; promotion vs attrition; peak shift with board size / Λ (CODA-owned).
Basketball (SCOUT)	Yes — this is where generative lives	Rung 1 LOO inverted-U (empirical) + Rung 2 Alex score POC (generative on pool mean).
Tenure (PEER)	Yes — tenure does not need generative sim for v1	Rung 1 only: preliminary empirical inverted-U on <code>poolq_loo_mean</code> (stage 9, ~55 depts inference N). Layer B Cox = pre-submission upgrade, not Tier 2 closure. <i>(PEER Round 2 confirmed.)</i>

Version B vs Version C (today):

- **Version B (qualitative cross-domain consistency): Satisfied** — Army established, basketball established, tenure **preliminary** (honest label). One shared phenomenon (inverted-U on LOO peer-quality proxy); setting-specific methods.
- **Version C (prediction test from minimal model): Partially satisfied** — basketball near-threshold heterogeneity (SCOUT); Army near-threshold + Λ -shift (CODA); tenure **not required** for v1. Basketball generative closes the **mechanism** leg; Army/tenure close **empirical** legs at different maturity.

Under Path II, Charles does **not** need to “apply the generative model” in all three domains — one generative proof-of-concept (basketball) + three empirical stylized facts (at varying maturity) is the v1 architecture.

§5 — Single next SCOUT task

Field	Answer
Task	Build <code>sports/scripts/export_scout_manuscript_bundle_v1.py</code> → output <code>datasets/mbb/exports_inverted_u_v0/scout_manuscript_v1/</code> (unless Charles locks C9 alternate path)
Unlocks	C1, C2, C3, C4, C7 → green ; refreshes empirical Fig 2 slug; freezes generative contrast PNGs (ability-only vs 539 preset; pool-mean + LOO-pool-quality readout panels); writes <code>manifest.json</code> , <code>axis_table_generative_readouts.md</code> , <code>score_equation_one_pager.md</code>
Effort	1–2 sessions — packaging + one <code>sports/530_sports_pipeline.ipynb</code> re-export; no new science
Charles locks needed first?	Y — C9 only (bundle directory path). C7–C8 (PERF_METRIC, bin count) already SCOUT-defaulted in <code>20260611_1640_SCOUT_to_COMPASS.md</code> ; proceed with defaults if Charles silent

§6 — Handoff to VECTOR

Ready to ink (now)

- Rung 1 empirical claim: inverted-U on **LOO pool quality** in college basketball (`530` / `538` notebooks).
- Path II architecture sentence: generative POC ≠ LOO-axis replication.
- Nesting chain from `20260611_1640_SCOUT_to_COMPASS_model_coherence.md` §B5.
- Limitation sentence from same file §A3 (Rung 2 vs Rung 1 axis).
- Talent-only failure as **concept** (mechanism needs congestion term).
- Setting 3 tenure = **preliminary** empirical leg (PEER stage 9).

Ink with caveat

- **Generative peak-and-decline figure** — caveat: *“Qualitative POC on whole-roster pool mean; not a reproduction of the LOO-pool-quality empirical axis.”*
- **Near-threshold heterogeneity figure** — caveat: *“Empirical prediction-facing readout; not fitted from generative simulation.”*
- **Score equation** — caveat: *“539 preset constants; not domain-calibrated λ.”*

Do not ink yet

- “Generative model reproduces basketball inverted-U on LOO pool quality.”
- “538D implements full B(Q)–D(Q) decomposition.”
- “Peak location shifts with Λ in basketball generative sim” (unless/until CODA+SCOUT joint export exists).
- Any June 2026 empirical Fig 2 caption without checking slug date in bundle manifest.

§7 — One sentence for Charles (Alex script)

“We will call the minimal model complete when the basketball generative score POC is frozen in a manuscript export bundle — talent-only fails, congestion-in-score bends curves on pool mean, at least one model-guided empirical quantity is exported (team quality vs viable-peer congestion), at least one prediction readout is on disk (near-threshold heterogeneity), axis table and limitation prose are frozen — while Army and tenure stay empirical inverted-U legs at honest maturity, without generative LOO bin-for-bin match or full 3-domain parameter identifiability.”

PD12 **B-lite** closure (COMPASS label, not Alex’s word) — see `14_PD12_reassessment_memo.md` Q2 and `#01` glossary.

Round 2 — SCOUT read receipt (all correspondence)

SCOUT read all agent-facing files in `3-Master_Plan/` (37 files). **Reading everything was helpful for this file’s §4** (CODA Army row, PEER tenure row) and for COMPASS stale-reference flags (April Fig 2 slug, PEER Cox wording).

New since Round 1	SCOUT action
<code>obsolete/correspondence_rounds/20260615_1012_PEER_to_SCOUT_round2.md</code>	Incorporated checklist §4 tenure row; bin-spec defer accepted
<code>20260615_1010_COMPASS_Round1_correspondence_audit.md</code>	Confirms 1626 round complete; this file closes June 15 queue
<code>20260615_1008_CODA_Round1_agent_mailbox.md</code>	Incorporated Army checklist §4 row
<code>20260615_1008_PEER_round1_inbox_and_cross_agent.md</code>	Read; aligned

SCOUT holds after this file unless Charles routes D10 implementation or VECTOR asks axis-table wording for tenure proxy row.

End SCOUT minimal model closure — Round 2.